

Digilians

Math Basics

Session 1: (Statistics Fundamentals)

Session Agenda

- ☐ Introduction to Statistics
- ☐ Descriptive Statistics

Introduction to Statistics

Importance of Statistics in AI and Machine Learning

Data Analysis

Statistics provide the tools to analyze and interpret data, which is the backbone of AI. Understanding data distributions, variability, and patterns is crucial for building robust models.

Model Evaluation

In AI, especially in machine learning, evaluating the performance of models (e.g., accuracy, precision, recall) relies on statistical measures.

Decision Making

AI models often make decisions based on probabilities. Understanding probability and statistical inference is key to trusting and explaining these decisions.

Real-World Applications

From predicting customer behavior to diagnosing diseases in healthcare, statistics provide the foundation for analyzing data and making informed decisions.

Basic Statistical Definitions

Population vs. Sample

Population

The entire group of individuals or items that you're interested in studying.

Example: If you want to study the average height of adult males in a country, the population would be all adult males in that country.

Sample

A subset of the population that is used to represent the whole.

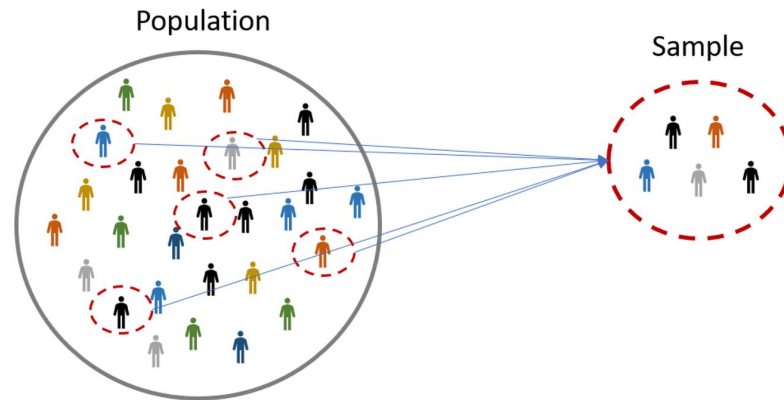
Example: Measuring the height of 1,000 adult males from different regions could be a sample representing the entire population.

Basic Statistical Definitions

Population vs. Sample

Population: The entire group of individuals or items that you're interested in studying. For example, if you want to study the average height of adult males in a country, the population would be **all** adult males in that country.

Sample: A subset of the population that is used to represent the whole. For instance, measuring the height of **1,000 adult males** from different regions could be a sample representing the entire population. Sampling is often done because it is impractical or impossible to study the whole population.

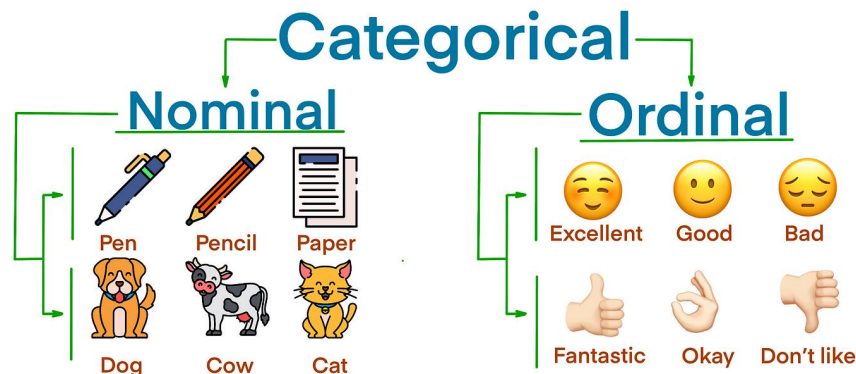


Types of Variables

Categorical (Qualitative) Variables

These variables represent categories or groups and can be either:

- **Nominal:** No inherent order among categories. Examples include colors (red, blue, green), gender (male, female).
- **Ordinal:** There is an inherent order, but the difference between levels is not quantifiable. Examples include customer satisfaction ratings (poor, fair, good, excellent), education levels (high school, bachelor's, master's).

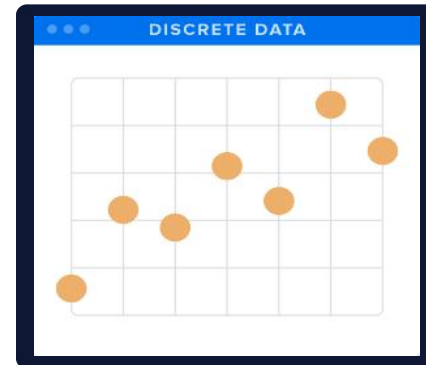


Types of Variables

Numerical (Quantitative) Variables

These variables represent measurable quantities and can be either:

- **Discrete:** Countable, finite values. Examples include the number of students in a class, the number of cars in a parking lot.
- **Continuous:** Infinite possibilities within a range. Examples include height, weight, temperature.



Real-World Relevance and Applications

Examples of Statistics in AI Applications

- **Fraud Detection:** Analyzing transaction data to identify patterns that indicate fraudulent activity. Here, statistics help to detect anomalies that deviate from normal behavior.
- **Recommendation Systems:** E-commerce platforms like Amazon use statistics to analyze customer preferences and recommend products based on what similar customers have purchased.
- **Healthcare Analytics:** Predicting the likelihood of diseases based on patient data. Statistical models can assess risk factors and provide personalized healthcare recommendations.



Descriptive Statistics

Descriptive Statistics

Descriptive statistics help provide a clear understanding of data through numerical calculations, graphs, and tables. This is a crucial step before conducting any further statistical analysis or building machine learning models.

Cases vs. Variables

Cases: A case represents an individual entity or subject in a dataset on which measurements or observations are made. It can be a person, a country, an event, or any other subject of study.

Variables: A variable is a characteristic or attribute that can be measured or observed for each case. Variables are features or properties that describe the cases.

		Variables			
		Store	Sales	Customers	Refunds
Cases	1	14	9	3	
	2	19	13	4	
	3	22	19	4	
	4	24	20	3	
	5	29	26	8	
	6	40	34	6	

Data Matrix vs. Frequency Table

Data Matrix

A data matrix is a structured table where each row represents a single case (individual data point), and each column represents a variable (characteristic or feature of the cases).

		Variables			
Cases	sepal length	sepal width	petal length	petal width	class
	5.1	3.5	1.4	0.2	Iris-setosa
	4.9	3	1.4	0.2	Iris-setosa
	6.5	3.2	5.1	2	Iris-virginica
	6.4	2.7	5.3	1.9	Iris-virginica
	6.8	3	5.5	2.1	Iris-virginica
	6.7	3.1	4.4	1.4	Iris-versicolor
	5.6	3	4.5	1.5	Iris-versicolor
	5.8	2.7	4.1	1	Iris-versicolor

Data Matrix vs. Frequency Table

Data Matrix

- **Purpose:** The data matrix is useful for raw data representation where each case's specific details are important. It's a comprehensive way to store all the information but not necessarily the best for summarizing data.
- **When to Use:** Data matrices are used in scenarios where you need to keep all details about each individual case, such as during data collection or when you need to perform case-by-case analysis.

Data Matrix vs. Frequency Table

Frequency Table

A frequency table is a summary of the data that shows how often each value of a variable occurs. It can display frequencies, percentages, and cumulative percentages.

Score	Frequency
50-59	2
60-69	2
70-79	6
80-89	7
90-99	3

Data Matrix vs. Frequency Table

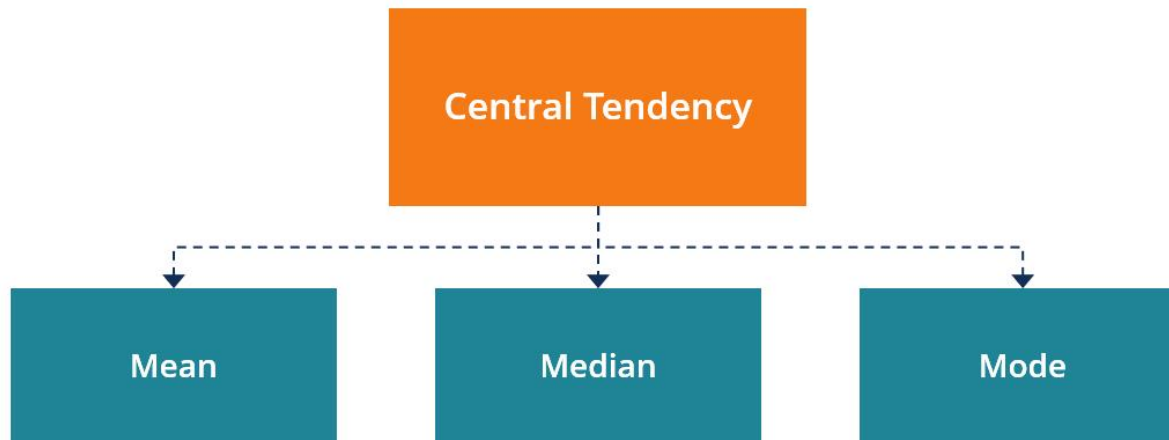
Frequency Table

- **Purpose:** Frequency tables are used to summarize and visualize the distribution of a variable, making it easier to see patterns and understand the data's structure.
- **When to Use:** When you want to quickly grasp how data is distributed across different categories or ranges, especially when dealing with categorical or quantitative variables.

Descriptive Statistics

Measures of Central Tendency

Understand how to describe the center of a dataset using different measures and learn how to calculate and interpret these values.



Measures of Central Tendency

Mean (Average)

The mean is the sum of all values divided by the number of values. It is a commonly used measure of central tendency.

Formula

For a dataset with values $x_1, x_2, \dots, x_n$, the mean μ is calculated as
$$\mu = \frac{\sum_{i=1}^n x_i}{n}$$

Example

For the dataset $[4, 8, 6, 5, 3, 4]$, the mean is
$$\mu = \frac{4 + 8 + 6 + 5 + 3 + 4}{6} = 5$$

The mean is sensitive to outliers. If the dataset has extreme values (very high or very low), the mean can be misleading

Measures of Central Tendency

Median

The median is the middle value of a dataset when it is ordered from smallest to largest. If the dataset has an even number of values, the median is the average of the two middle values.

Example

For the dataset [4,8,6,5,3,4], first sort the data:[3,4,4,5,6,8]. The median is the average of the two middle numbers, 4 and 5, which is $\frac{4+5}{2} = 4.5$

The median is not affected by outliers and provides a better measure of central tendency for skewed distributions.

Measures of Central Tendency

Mode

The mode is the value that appears most frequently in a dataset.

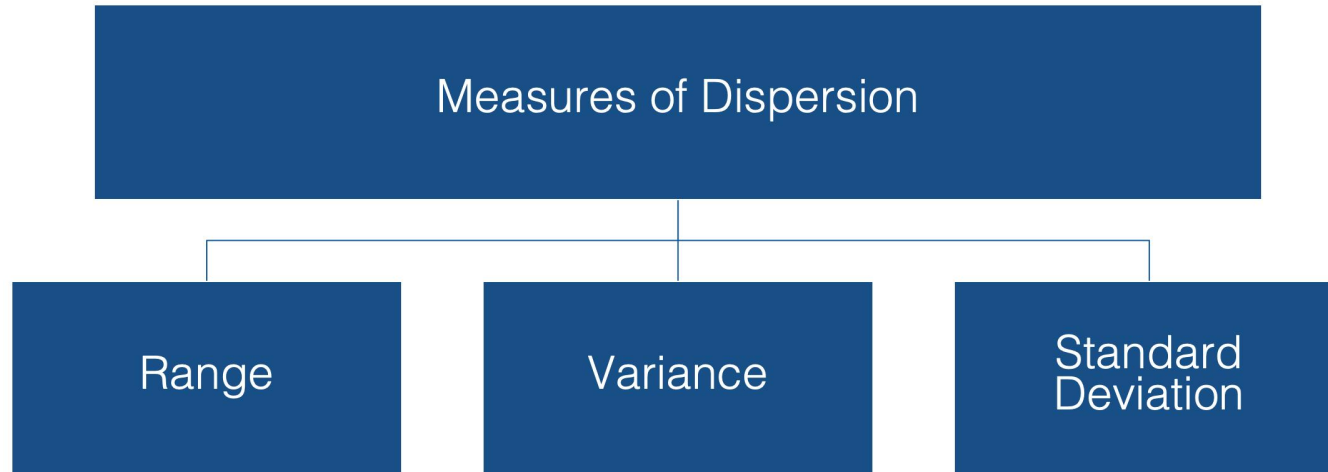
Example

In the dataset [4,8,6,5,3,4], the mode is 4 because it appears twice, more than any other number.

The mode is useful for categorical data and for identifying the most common item in a dataset. A dataset can have more than one mode (bimodal or multimodal) or no mode at all.

Measures of Dispersion

Learn how to describe the spread or variability of a dataset using different measures.



Measures of Dispersion

Range

The range is the difference between the maximum and minimum values in a dataset.

Formula

Range = Max value – Min value.

Example

For the dataset [4,8,6,5,3,4], the range is $8-3=5$.

The range gives a quick sense of the spread of the data but is highly sensitive to outliers.

Measures of Dispersion

Interquartile Range (IQR)

The IQR is the range between the first quartile (Q_1) and the third quartile (Q_3). It measures the spread of the middle 50% of the data, effectively capturing the central tendency without being influenced by extreme values or outliers. The IQR is calculated as:

$$IQR = Q_3 - Q_1$$

- Q_1 (First Quartile): The value below which 25% of the data falls.
- Q_3 (Third Quartile): The value below which 75% of the data falls.

Interquartile Range (IQR)

Why Use IQR?

- **Robustness Against Outliers:** Unlike the range, which considers only the minimum and maximum values, the IQR focuses on the middle 50% of the data. This makes it less sensitive to outliers and extreme values, providing a more reliable measure of dispersion for skewed distributions.
- **Understanding Data Spread:** The IQR helps understand the variability of the central portion of the data. A larger IQR indicates more spread in the middle 50% of the dataset, while a smaller IQR suggests that the data points are closer to the median.

Interquartile Range (IQR)

Example Calculation of IQR

Let's use a small dataset to illustrate: Dataset: [2,4,4,5,6,8,9]

1. Arrange the data in ascending order (already sorted in this case).
2. Find Q1 and Q3:
 - **Q1 (First Quartile):** The median of the first half of the dataset (excluding the median if the number of data points is odd). For [2,4,4], $Q1 = 4$
 - **Q3 (Third Quartile):** The median of the second half of the dataset. For [6,8,9], $Q3 = 8$
3. Calculate IQR:

$$IQR = Q3 - Q1 = 8 - 4 = 4$$

Measures of Dispersion

Variance

Variance measures the average squared deviation of each number from the mean. It provides insight into the spread of all data points around the mean.

Formula

For a dataset with values $x_1, x_2, \dots, x_n$, the variance σ^2 is calculated as
$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n}$$

Example

For the dataset [4,8,6,5,3,4], calculate the mean ($\mu=5$), then compute the variance:

$$\sigma^2 = \frac{(4-5)^2 + (8-5)^2 + (6-5)^2 + (5-5)^2 + (3-5)^2 + (4-5)^2}{6} = \frac{1+9+1+0+4+1}{6} = 2.67$$

Variance is in squared units, making it less interpretable in the original scale. However, it is fundamental in statistical theory.

Measures of Dispersion

Standard Deviation

The standard deviation is the square root of the variance and provides a measure of the average distance from the mean. It is in the same units as the data, making it more interpretable.

Formula

$$\sigma = \sqrt{\sigma^2}$$

Example

For the dataset [4,8,6,5,3,4], the standard deviation is: $\sigma = \sqrt{2.67} \approx 1.63$

A small standard deviation indicates that the values are close to the mean, while a large standard deviation indicates that the values are spread out over a wider range.

Descriptive Statistics

Data Distribution and Visualization

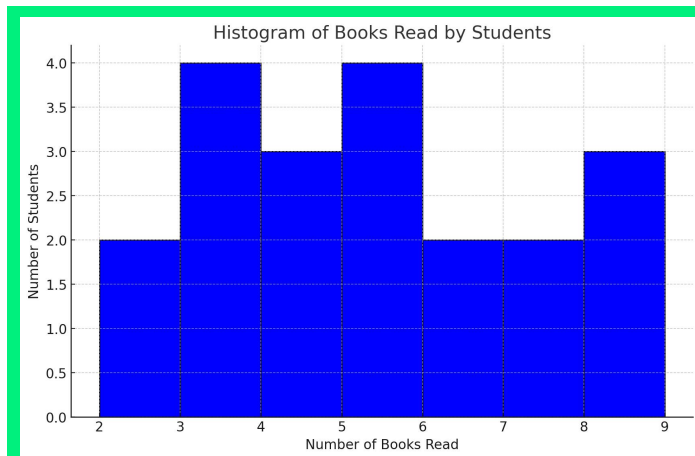
Learn how to visualize data distribution and understand its shape
using graphical representations.

Descriptive Statistics

Data Distribution and Visualization

Histograms: A histogram is a graphical representation that organizes a group of data points into user-specified ranges. It shows the frequency distribution of a dataset.

Example: Create a histogram for a dataset representing the number of books read by students in a class.



Histograms help visualize the shape of the data distribution (e.g., normal, skewed) and identify patterns like bimodality or skewness.

Descriptive Statistics

Data Distribution and Visualization

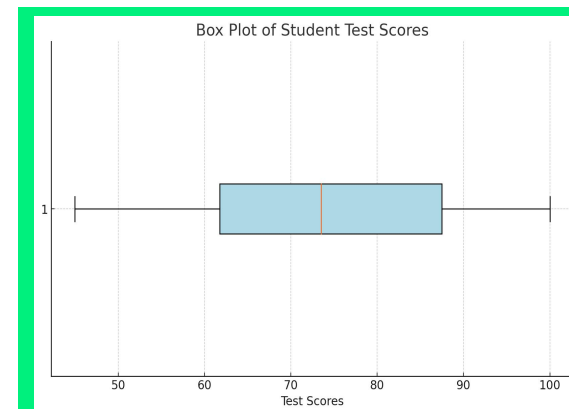
Box Plots (Box-and-Whisker Plots): A box plot displays the distribution of data based on a five-number summary: minimum, first quartile (Q1), median (Q2), third quartile (Q3), and maximum.

Example: Create a box plot for a dataset of test scores to identify the median score, quartiles, and potential outliers.

Quartiles:

First Quartile (Q1, 25th percentile): The left edge of the box represents the 25th percentile, indicating that 25% of the students scored below this value.

Third Quartile (Q3, 75th percentile): The right edge of the box represents the 75th percentile, showing that 75% of the students scored below this value.



Box plots are useful for detecting outliers and understanding the spread and symmetry of the data.

THANK
you